Chapter 6

6.1 Basics of Work Health and Safety

6.1.1 Safe and healthy work environment

The workspaces and production methods, in a good working environment, have been designed and implemented in such a way that workers can work and move around safely. Workers are familiar with the hazards and risks related to the raw materials used in the work and the substances produced in the work processes. They are well trained to control them efficiently. The machines and tools used in the work suit their purposes. Employees' physical and mental preconditions are taken into account while planning and scaling the work.

Workspaces and Passageways in a Safe Environment

In a safe working environment:

- Structural aspects of the work environment take into account the safety of passageways, workplace lighting, sound environment and indoor air quality.
- Functional factors include the organization of transportation and keeping workspaces and offices organized and neat.

Order and cleanliness

Cleanliness contributes to the safety and health of the workplace.Therefore, it is a top priority. Slipping and tripping caused by the disorder or untidiness of the workplace make up a significant portion of workplace accidents. Special consideration must be paid to the identification and management of physical, chemical and biological health hazards in the workplace. Work machinery, equipment and tools must be in good working order, and they may be used only for the planned work and under the intended conditions. The necessary personal protective equipment and assistive devices must also be in good working order and used only for the planned situation.

Fig 6.1 Example of a good working environment

6.1.2 Rules and Principles of Work Health and Safety

- All people are given the highest level of health and safety protection that is reasonably practicable.
- Those who manage or control activities that give rise, or may give rise, to risks to health or safety are responsible for eliminating or reducing health and safety risks.
- Employers and self-employed people should be proactive and take reasonably practicable measures to ensure health and safety in their business activities.
- Employers and employees should exchange information about risks to health and safety and measures that can be taken to reduce those risks.
- Employees are entitled, and should be encouraged, to be represented on health and safety issues.

6.1.3 Ergonomics

Ergonomics is the process of designing or arranging workplaces, products and systems so that they fit the people who use them. Ergonomics applies to the design of anything that involves people – workspaces, sports and leisure, health and safety. Itis a branch of science that aims to learn about human abilities and limitations, and then apply this learning to improve people's interaction with products, systems and environments.

Ergonomics helps to improve workspaces and environments to minimize risk of injury or harm. Ergonomics should be applied to design amenities of a working environment. As technologies change, so does the need to ensure that the tools we access for work, rest and play are designed for our body's requirements.

Fig 6.1 Ergonomically Designed Chair

6.1.4 Obligations and Ethics towards WHS

Some simple steps to remember when considering ethics in WHS profession are:

- Follow the governmental requirements, laws, guidelines, as well as company principles and policies when conducting WHS related tasks and making decisions.
- Be transparent in communicating data, findings, recommendations to the respective management.
- Follow the code of ethics of the relevant professional association and/or accreditation body and report any incident
- Detach personal judgement, thoughts and ideas from professional setting as much as possible.
- Support and follow WHS values and principles

6.1.5 Common Workplace Hazards

The six main categories of hazards are:

- **Biological**: Biological hazards include viruses, bacteria, insects, animals, etc., that can cause adverse health impacts. For example blood and other bodily fluids, harmful plants, sewage, dust and vermin.
- **Chemical**: Chemical hazards are hazardous substances that can cause harm. These hazards can result in both health and physical impacts, such as skin irritation, respiratory system irritation, blindness, corrosion and explosions.
- **Physical**: Physical hazards are environmental factors that can harm an employee without necessarily touching them, including heights, noise, radiation and pressure.
- **Safety**: These are hazards that create unsafe working conditions. For example, exposed wires or a damaged carpet might result in a tripping hazard. These are sometimes included under the category of physical hazards.

- **Ergonomic**: Ergonomic hazards are a result of physical factors that can result in musculoskeletal injuries. For example, a poor workstation setup in an office, poor posture and manual handling.
- **Psychosocial**: Psychosocial hazards include those that can have an adverse effect on an employee's mental health or wellbeing. For example, sexual harassment, victimization, stress and workplace violence.

6.1.6 Managing Workplace Hazards

If it is not possible to eliminate the hazard.Below are six steps to determine the most effective measures to control workplace hazards and to minimise risk.

Step 1: Design or re-organise to eliminate hazards.

Step 2: Substitute the hazard with something safer.

Step 3: Isolate the hazard from people.

Step 4: Use engineering controls.

Step 5: Use administrative controls.

Step 6: Use Personal Protective Equipment (PPE).

6.1.7 Purpose of Organizational Policies, Procedures and Processes

The purpose of organizational policies, procedures, processes and systems for WHS are to establish the practices and standards that a company will follow in regards to compliance with Work Health & Safety guidelines.

- These company policies and procedures will ensure that the company is in full compliance with the legislative requirements concerning work health & safety for employees
- The policies will create a safe work environment for all employees and will reduce the cost the company has for sending employees to see doctors after accidents, and will reduce the insurance costs of the company because of their safety rating

- The establishment of the policies will create better work relationships between company owners, and their employees. The employees will know that the company is trying to protect them from injury

6.2 WHS in IoT Environments

6.2.1 Understand the Scope of Project

IoT working environments demand extreme precautions while handing IoT equipment. The first and foremost requirement is to understand the scope of the project i.e. to understand the product development cycle. For instance what processes will be involved in designing an IoT product or project. Some IoT projects will be hardware based, some will be software based. So, WHS requirements will vary for different IoT projects.

6.2.2 Risk Profile for different Stakeholders

Different teams are involved in the development of an IoT system i.e., software team, electronics team, PCB team and QA etc. So, WHS requirements will vary for the different teams. For example, WHS principle for software team will be different as compared to that of hardware team.

6.2.3 Client's Health and Safety SOPs

There is always a need to develop SOPs to operate the IoT product or system. These SOPs will be helpful for the client's health and safety. IoT devices have different contexts and require separate SOPs for each application. For example, smart homes and smart cities will require different SOPs for the smooth and safe operation of an IoT system. For instance, an IoT sensor device which is installed in the pantry should have strict SOPs regarding the installation and operation of device. SOPs will be such as: The IoT device installed in the pantry should be at a defineddistance from the hearth, and it should not be installed in the path of smoke etc.

6.2.4 Development Techniques of Health and Safety Plan

Development of health and safety plan in IoT working environment is directly related to the design of IoT devices i.e., what is the maximum temperature the device can bear; Water resistivity of the device; working temperature range of components (sensors and actuators) installed in the device; and operating current and voltage of the device etc. All these design parameters will lead to a successful health and safety plan.

Fig 6.3Example of a good IoT enabled environment

6.2.4 ImplementationTechniques of Health and Safety Plan

Nowadays IoT devices are being used to monitor work health and safety SOPs. To better implement the health and safety plan in an IoT working environment, some IoT devices must be dedicated to sense and report the cases where SOPs are not followed. For instance, tools in an IoT working environment can be synchronized with the personal protective equipment. Tools will not turn on unless the personal protective equipment is worn. Figure 6.1 shows an example of intelligent personal protective equipment.

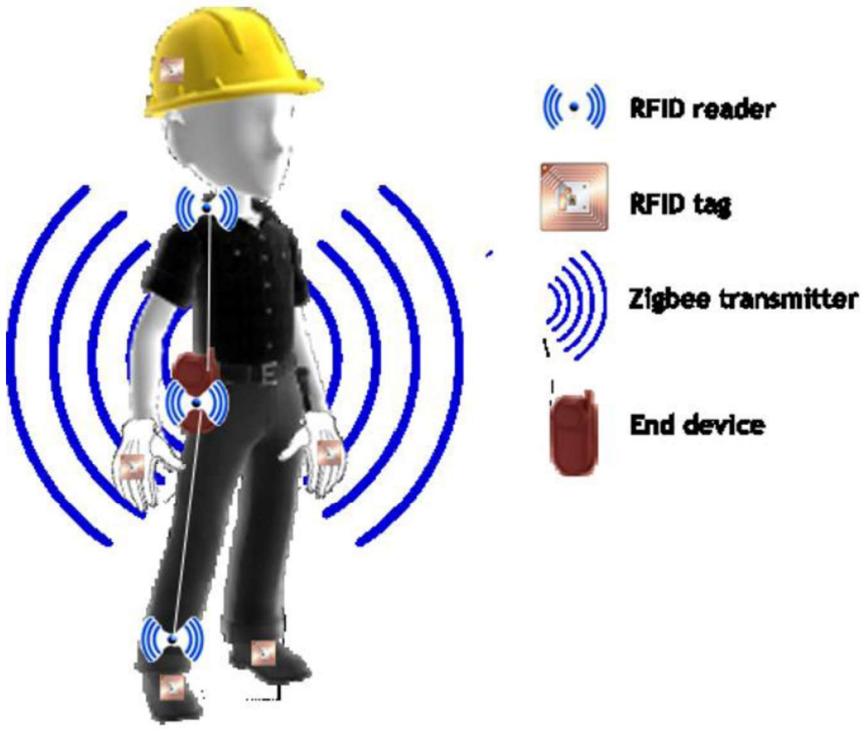

Fig 6.4IoT based protective equipment

Key Points

- Structural aspects of the work environment should take into account the safety of passageways, workplace lighting, sound environment and indoor air quality.
- Functional factors of the work environment must include the organization of transportation and keeping workplaces and offices organized and neat.
- Those who manage or control activities that give rise, or may give rise, to risks to health or safety are responsible for eliminating or reducing health and safety risks.
- Employers and employees should exchange information about risks to health and safety and measures that can be taken to reduce those risks.
- Psychosocial hazards include those that can have an adverse effect on an employee's mental health or wellbeing.

Exercise

Write short answer of the following

1. Define ergonomics.
2. Describe electrical hazards.
3. Enlist hazards involved in an electronics work environment.
4. What are the hazards involved with the handling of IoT devices?

Answer the following question in detail.

1. Describe basic rules and principles of WHS.
2. Describe Ergonomics.
3. What are common workplace hazards?
4. How workplace hazards should be managed?
5. What are the WHS requirements for IoT workplaces?
6. What are the implementation techniques for WHS in IoT environments?